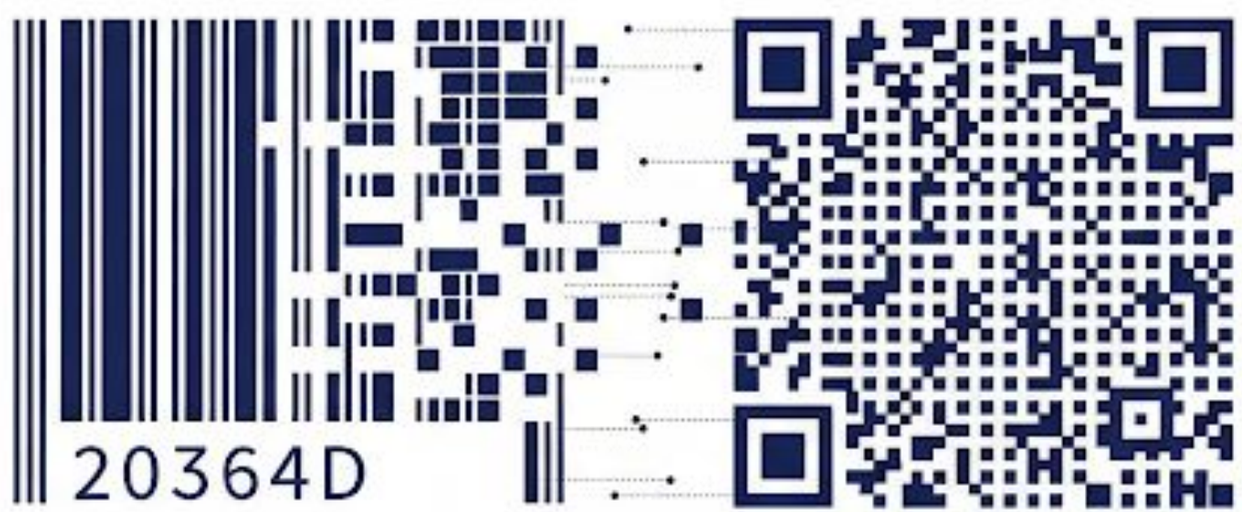
Digital Product Passports & Apparel

@Closing the Loop,
St-Gallen, Switzerland

27.10.2026
Dom Guinard
Spotd.com



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About spotd.com

Strategic consultancy for connected products, made in Switzerland 🇨🇭

4+ Standards Co-Authored
GS1 Digital Link, EPCIS, etc.

30+ Projects Delivered
Ralph Lauren, Patagonia, Unilever, etc.

1B+ Products Connected
CPG, Apparel, Pharma

Who We Serve

Service Providers

scantrust:
DIGIMARC

Strategic & technical advisory for next-gen platform providers

Brands & Enterprises

PUTCH

Digital native products & regulatory compliance (DPP)

Private Equity & VC

Technical due diligence for digital identity investments

DPP Regulatory Framework

Understanding the regulatory maze





ESPR will deliver benefits for consumers, businesses and the planet

- ESPR products cover ca. **16% of the EU's GHG** emission
- Measures could **cut total EU GHG** emissions by up to **7%**
- Improve the EU's **industrial competitiveness** and **boost demand** for circular and sustainable products
- **Empower businesses and consumers** to make more sustainable products choices, based on **transparent product information**
- **Harness public purchasing power** (around **14% of EU GDP**, or approx. EUR 1,8 trillion annually) through green public procurement
- **Improve product performance** and **lifetimes** generating **financial savings** for consumers
- Increase **strategic autonomy** and **reduce raw material dependencies**

European Commission

DPP: A Digital Passport for all Products in the EU



Material suppliers

- Track raw materials extraction and production
- Support due diligence and material footprint tracking



Manufacturers

- Streamline documentation via product digital twins
- Centralize required information in one location



Market surveillance authorities

- Access traceable and verifiable documents (e.g., customs)
- Support surveillance and inspection processes



Public authorities & policy makers

- Access reliable product sustainability data
- Facilitate sustainability initiatives and policy proposals



Consumers

- Enhance transparency and verified info access
- Enable sustainable purchase options



Others (Recyclers & Repairers)

- Track full product lifecycle journey
- Enable new service models for repairers & recyclers

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Anatomy of an EU DPP

The four fundamental architectural components defining a Digital Product Passport

01

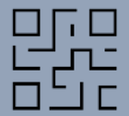


Identity

Key Elements

- Unique Product ID (UID)
- Operator / Facility ID
- Model & Batch Granularity

02



Data Carrier aka Tag

Key Elements

- QR Code / DataMatrix
- RFID / NFC Tags
- On-product Accessibility

03



Meta & Life-Cycle Data

Key Elements

- Material Traceability
- Carbon & Water Footprint
- Repair & Recycling Guides

04



Access Rights & APIs

Key Elements

- Role-Based Access (RBAC)
- Public vs Private Data
- Open Standards & APIs

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DPP Is Not Just One Document & Workstream

ESPR Framework (Ratified, May 2024)
Core requirements applying to ALL product categories

ESPR Delegated Acts

Category-Specific Requirements:
Identifier granularity, required data attributes, and verification methods per product type.

- **Batteries**
  2024 - 2027
- **Apparel**
  2027 - 2029
- **Next Categories**
  (?)

Other Regulations referencing DPP

- **Packaging Regulation (PPWR)**
- **Construction Products (CPR)**
- **Other Future Regulations**

CENELEC JTC 24 Technical Standards

ISO JTC 5 Technical Standards - Internationalization

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DPP Technical Details - What is Known Today from the ESPR

Identity

- Products will require a **unique identity** (at least SKU level) linking to a DPP [Art 9]
- Identity based on **recognized standard** [Art 9]

Carrier

- One or more data carriers leading to passport [Art 9]
- Data carrier needs to be included **on the product** [Art 9]
- Primary data carrier needs to use a **recognized standard** [Art 9]

Data

- Foster greener choices [Art 7]
- Substances of concern [Art 5]
- Carbon & Environmental Footprint [Art 5]
- Needs to remain available if a company disappears [Art 9]

Access

- **DPP Web &** easily accessible [Art 7]
- Data accessible & transferable via **APIs** [An 28]
- **Decentralized** data needs to be supported [An 32]
- Allowing differentiated & **granular access [Art 10]**

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CEN/CENELEC JTC 24 DPP Technical Standards

Module 1: 18219

Unique identifiers: ISO15459, GS1 Digital Link
https://dlnkd.tn.gg/01/952034541891 56/10/1234

Module 2: 18220

Data carriers: QR, Datamatrix, RAIN RFID, NFC

Module 3: 18239

Access rights management, information security

Module 4: 18223

System interoperability

Module 5: 18216

Data exchange protocols & data formats
JSON/LD, HTTP / REST

Module 6: 18221

Data storage, archiving, and data persistence
Backup

Module 7: 18246

Data Authentication, Reliability, and Integrity
Payload encryption

Module 8: 18222

Application Programming Interfaces (APIs) for the product passport lifecycle
CRUD APIs

4 Technical Layers of DPP



Trust Layer
How can the data be trusted?

e.g., DID & VC, Blockchain anchoring

Out of scope for now



Data Layer
What data format for the product passport?

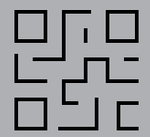
Format, storage - e.g., GS1-Web Vocabulary, JSON-LD, EPCIS, etc.

Format defined by JTC 24 standards, content defined by delegated acts



Identity Layer
What digital product identity is encoded the tag?

GS1 Digital Link,
ISO 15459



Carrier Layer
What is the on-product tag?

QR / Datamatrix,
RAIN RFID (UHF), NFC

Defined by JTC 24 standards

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Anatomy of a Delegated Act: Detergents

Annex VI

DIGITAL PRODUCT PASSPORT

PART A

Information to be included in the digital product passport

The ***digital*** product passport shall include the following information:

(a) the ***trade name,*** unique product identifier of the detergent or ***end-user*** surfactant ***and a colour image of the packaging or the label of the model of detergent or end-user surfactant of sufficient clarity to enable the identification***;

(b) the name, the ***postal and electronic*** address ***and telephone number*** of the manufacturer***, and, where applicable, of the importer*** or the ▌ authorised representative, as well the manufacturer's unique operator identifier;

(ba) the reference to the digital product passport service provider hosting the back-up copy of the digital product passport:

(c) the identification of detergent or ***end-user*** surfactant allowing traceability***;***

(ca) the mention that the digital product passport is issued under the sole responsibility of the manufacturer;

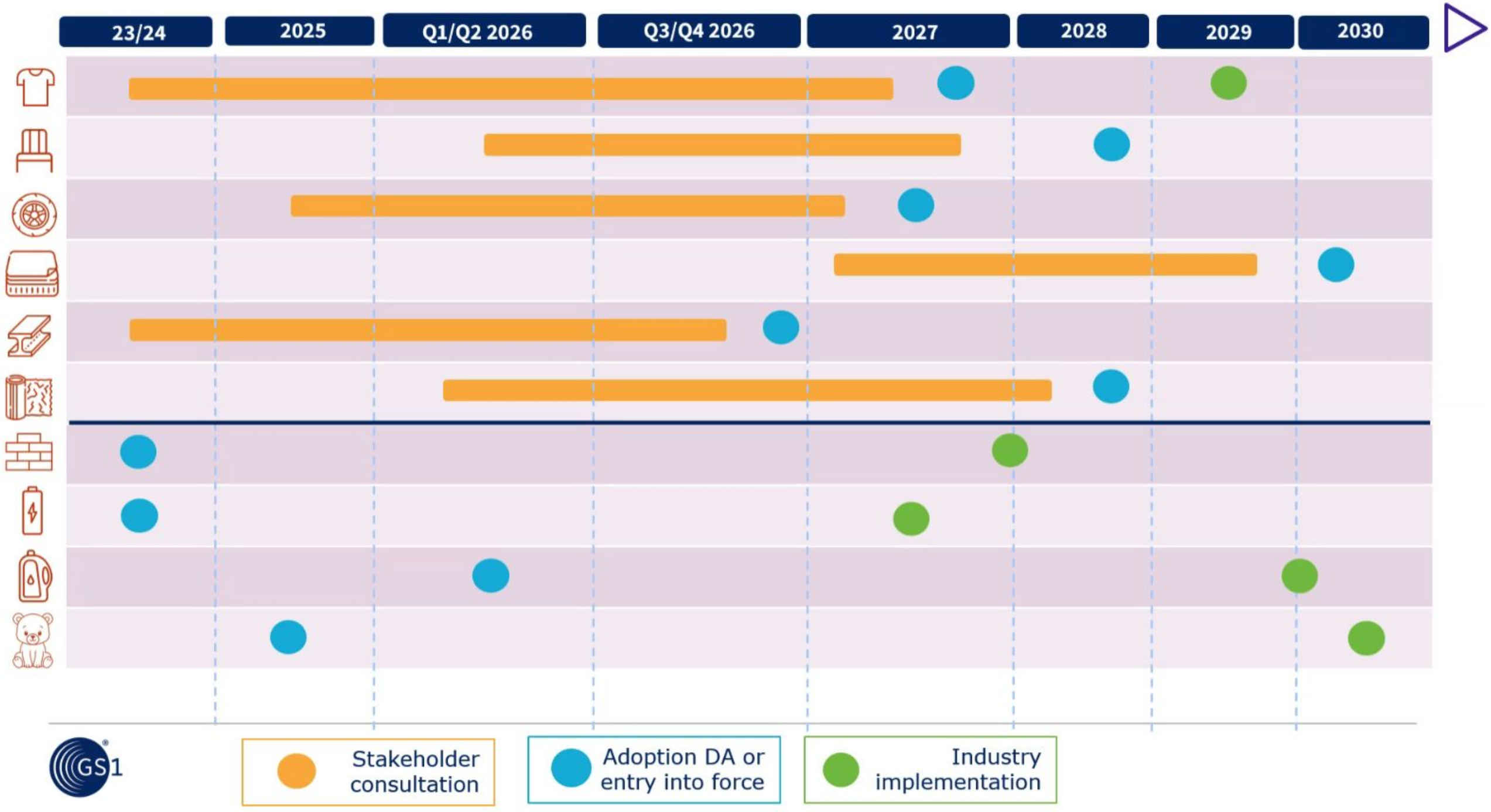
(d) ***where applicable,*** the commodity ***codes, as defined in Regulation (EEC) No 2658/87*** under which the detergent or ***end-user*** surfactant is classified at the moment the ***digital*** product passport is created***;***

(e) ***a statement that compliance of*** the detergent or ***end-user surfactant with this Regulation has been demonstrated, and, where relevant, references to other Union law with which the detergent or end-user*** surfactant complies ▌ ;

(f) a full list of substances intentionally added in the detergent or surfactant***, identified*** in accordance with ***Article 18(3) of Regulation (EC) No 1272/2008; carry-over preservatives shall also be listed if they must be included in the label in accordance with Article 18(3),*** point ***b, of Regulation (EC) No 1272/2008, Article 58 of Regulation (EU) No 528/2012 or point h(iv) of paragraph 1 of Part A*** of Annex V***;***

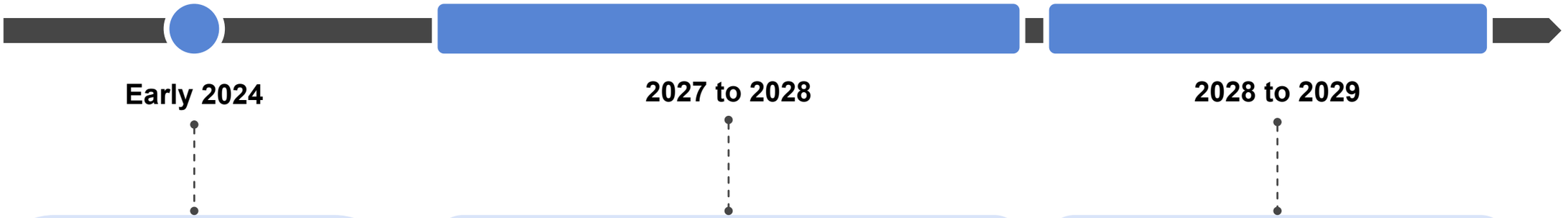
(g) the list of all intentionally added micro-organisms providing their taxonomic classification (genus, species and strain name or code).

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23/24
2025
Q1/Q2 2026
Q3/Q4 2026
2027
2028
2029
2030
Stakeholder consultation
Adoption DA or entry into force
Industry implementation

DPP For Apparel: Entering the Hotphase

Timeline of DPP legislation and implementation by European Commission



Early 2024

- May 2024 - core ESPR legislation adopted
- First priority product categories identified: Apparel, Detergents, Furniture, Mattresses Chemicals, Iron & Steel, Construction products, etc.

2027 to 2028

- 2027 - Publication of apparel delegated act

2028 to 2029

2027 - DPP becomes mandatory for apparel products

Key takeaways:
2027-2029 is likely is the hot phase for apparel: brands will need to get ready for DPP!

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Apparel Roll-Outs

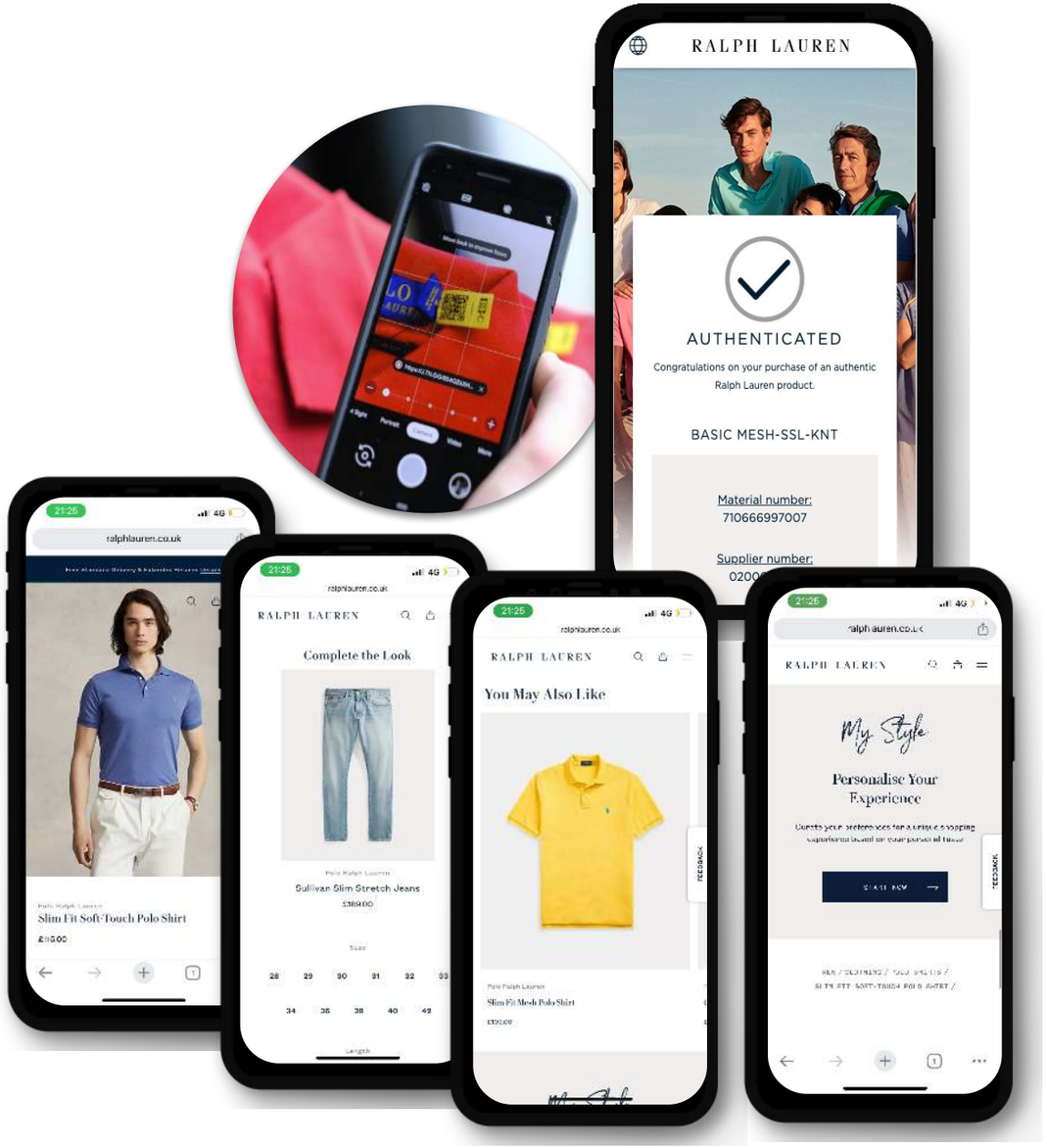
Pre-DPPs in the real-world



Ralph Lauren: Revenue Growth & Brand Integrity

Solution Overview

- GS1 Digital Link serialized hang tags
- Serialized sewn in tags
- 500 million+ products
- Born digital in the factories
- Authentication, customers loyalty, in-store experiences, etc.



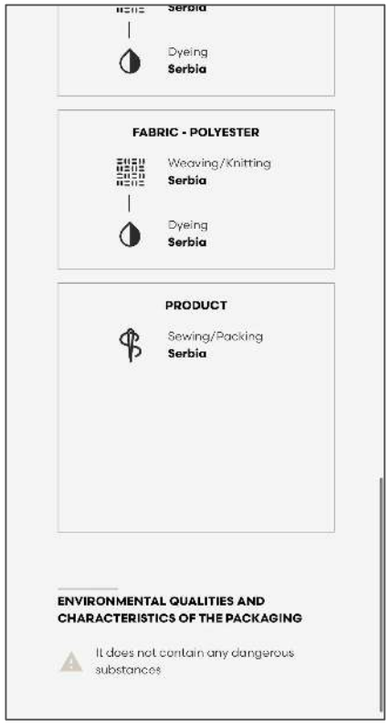
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Calzedonia: Preparing for DPP

Solution Overview

- SKU/batch and factory level QR codes
- QR code on care label & hang tags
- Data uploaded via PIM and via CSV files



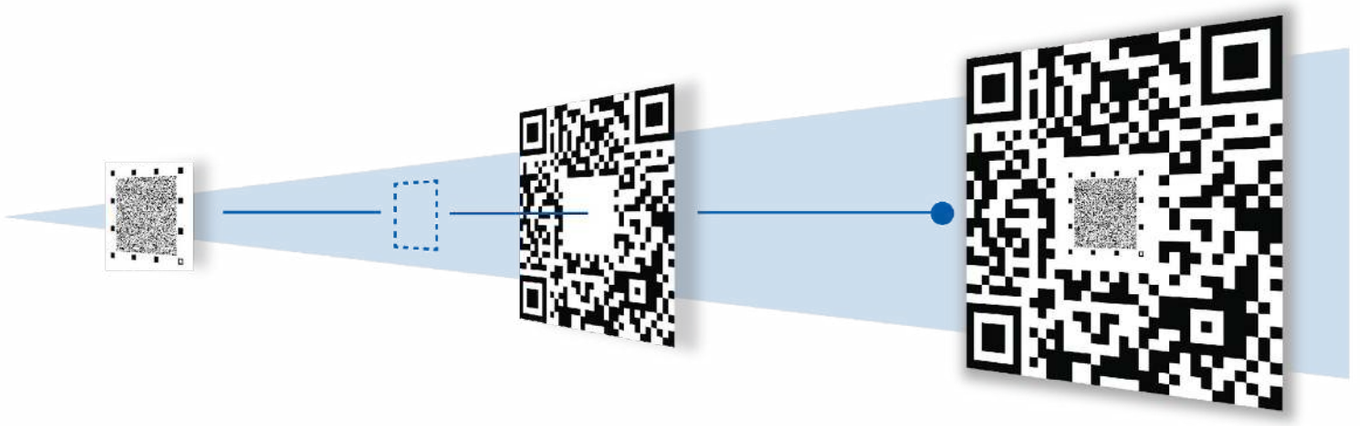




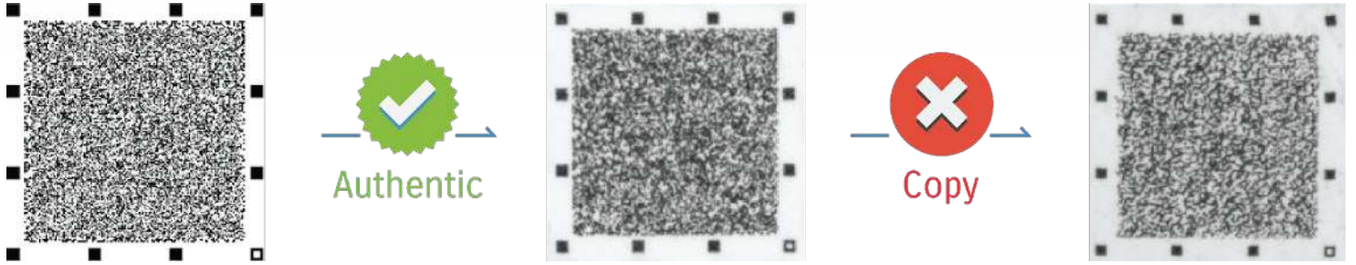
Try a sample "DPP lite" experience by scanning this code:

Anti-counterfeiting Benefits: World's First QR Code with Copy Detection

scantrust:



Secure Graphic + QR Code = Scantrust Secure Code



DIGITAL FILE
100% information quality

ORIGINAL PRINT
60% information quality

PHOTO COPY
30% information quality

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Patagonia: Preparing for DPP: Sustainability & Efficiency

Solution Overview

- GS1 Digital Link Serialized hang tags
- 40M single use hang tags and inserts saved from landfills
- 80 tons of paper saved annually
- Moving into serialized sewn in tags

Time & Efficiency: Hours Spent Annually Pre-Digitisation

348 Creative Management	**200** Design Team	**100** Design Team	**96** Product Line Manager
300 Proofing Team	**120** Production Team	**300** Packaging & Branding	**1,355** Corrections

Total Workdays Annually

353

2018

20 Cover Tag Styles

453 Insert Variations

2023

2 Cover Tag Styles

4 Auxiliary Tags



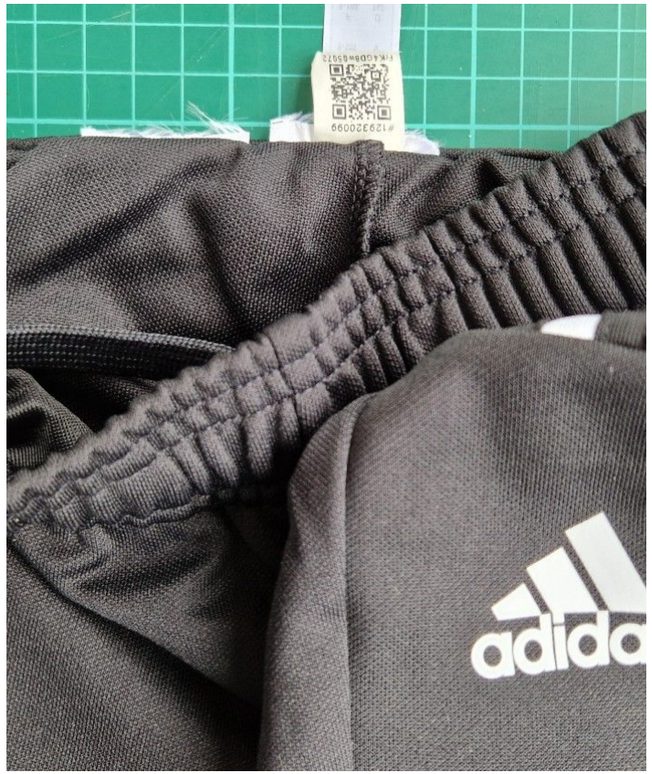
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A few other real world examples

Apparel

Adidas

Serialized QR: product information



Footwear

On Running

QR Code + UHF tag: product information, care information, traceability



Footwear

Puma

Serialized QR code: product information

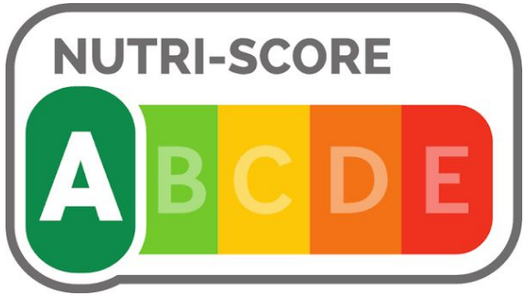


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“Brands should see DPP as an *opportunity*, not a legal burden. Focus on delivering *real* value for customers. It’s a unique opportunity to connect 121 with them!”







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Think outside the box...



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Thanks!

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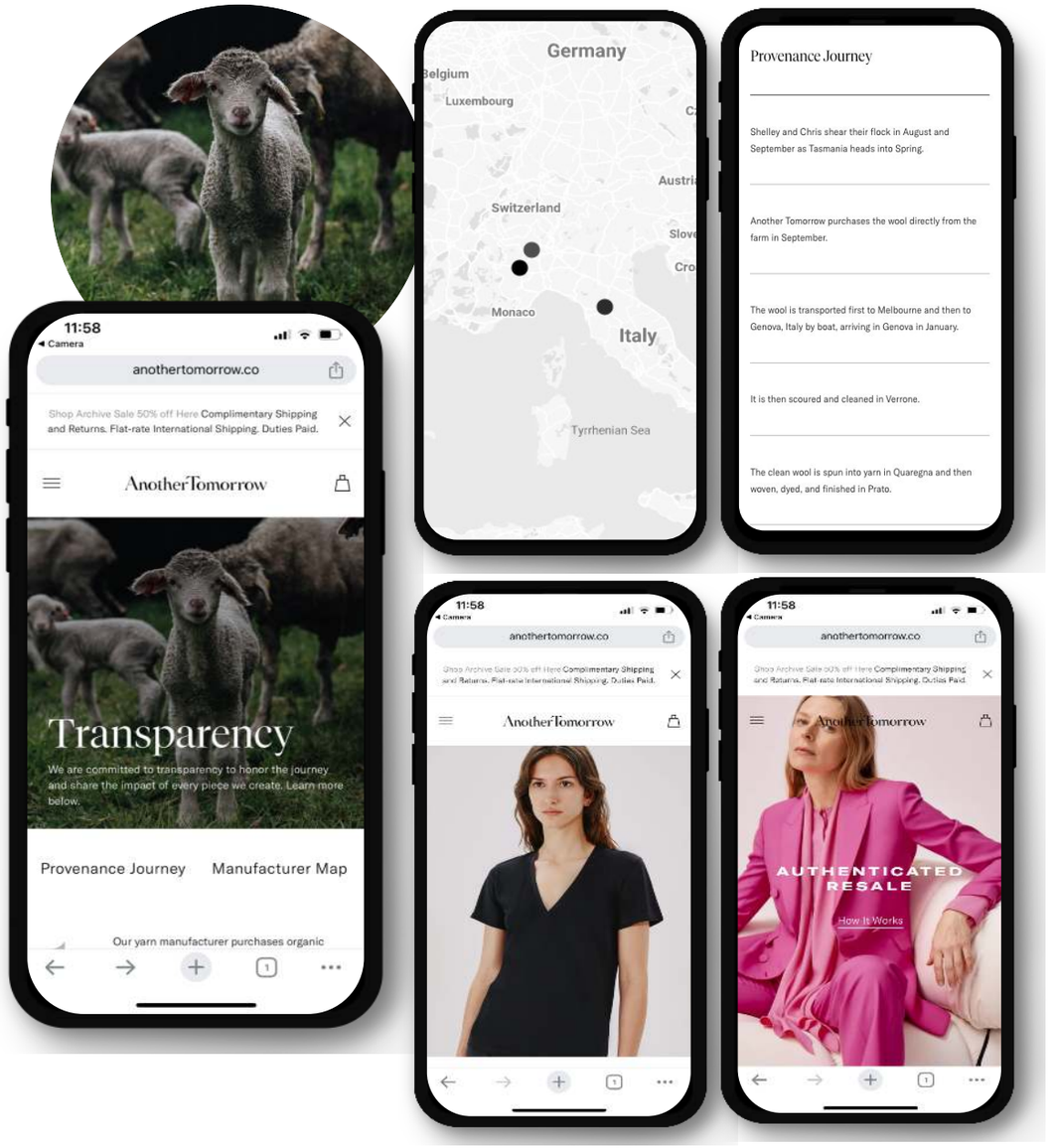
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digital identity consulting

Bringing transparency & sustainability to the fashion industry

Solution Overview

- Digitised garments deliver sustainability, brand purpose and transparency information to meet evolving consumer values
- Traceability data from origin enabling a rich digital consumer experience
- Building the foundation for a circular economy - digitising products to enabling more sustainable fibre



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